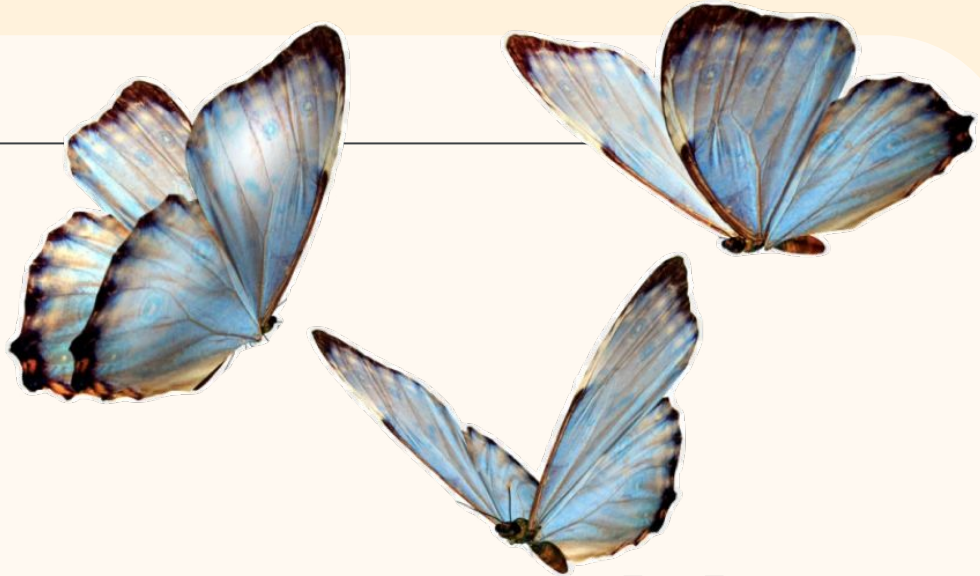
BIS15L



Temperature vs Nutrition

Perfecting larvae success

By GROUP 14

Why we chose the data?

- Butterflies contribute greatly to the diversity of plant species through cross-pollination
- Many crops require a pollinator such as a butterfly to reproduce
- By finding predicting factors, we could possibly estimate the behavior of butterflies given specific environmental conditions and support the protection of butterfly habitats based on these important factors

What stood out to us in the paper?

1 Importance of **plant nutrients**

Through a range of different variables including temperature, nitrogen levels among others. We saw in the paper that perfecting these levels leads to healthier plants which should be plants more attractive for butterflies to lay their eggs on.

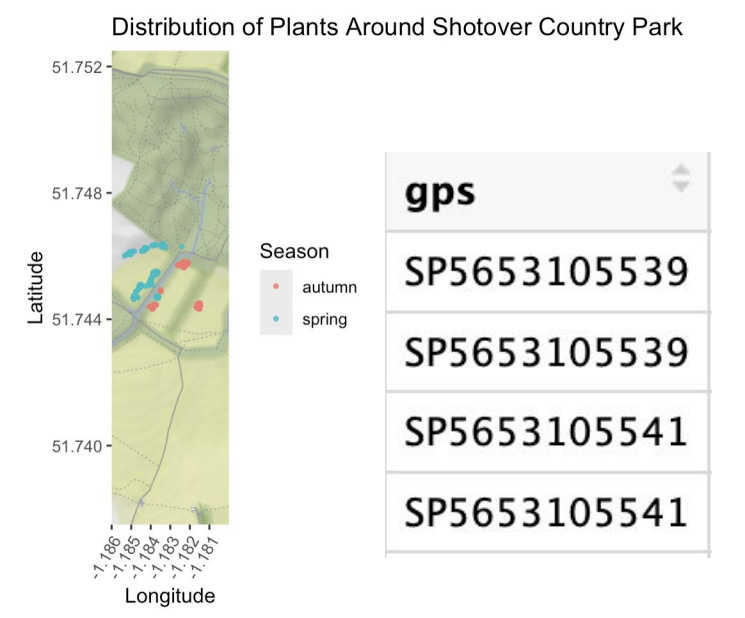
2 **Seasonal** Differences

The paper explores a range of factors including egg laying, plant health among other factors separately between the two seasons of spring and autumn.

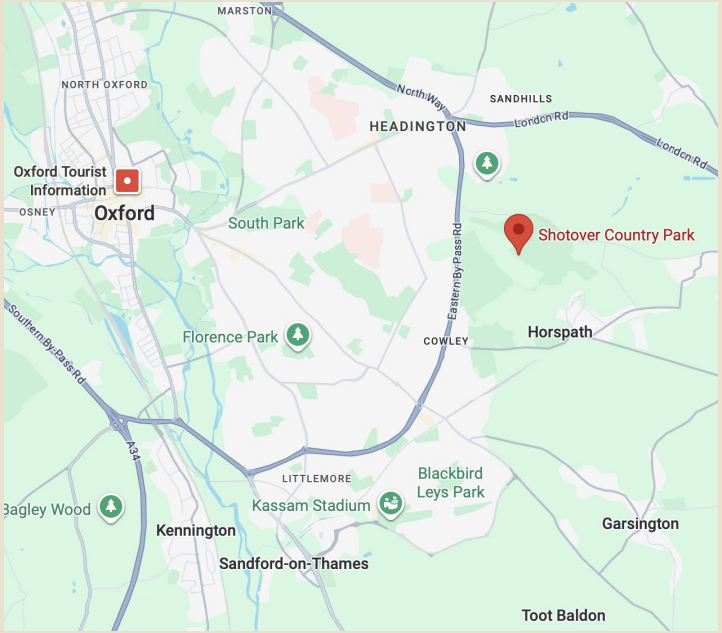
3 What do **butterflies** look for?

The paper looks at when and on what plants butterflies are laying their eggs to what season which is something we want to explore further in this presentation.

Map of plant distribution

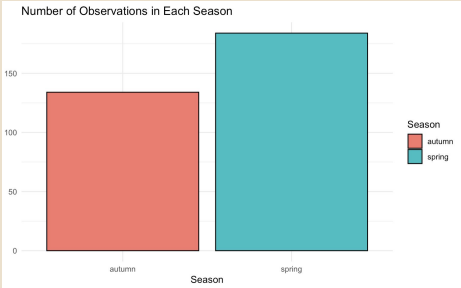


gps
SP5653105539
SP5653105539
SP5653105541
SP5653105541

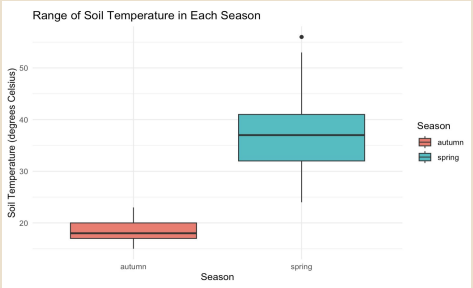


Exploratory analysis

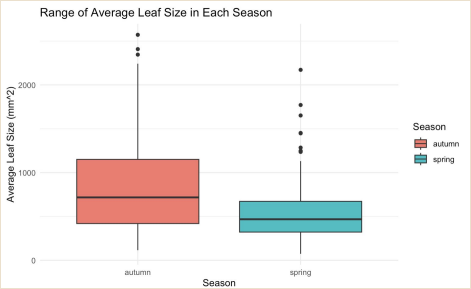
Counts of Each Season



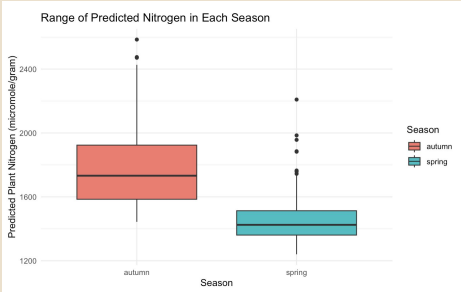
Soil Temperature Range



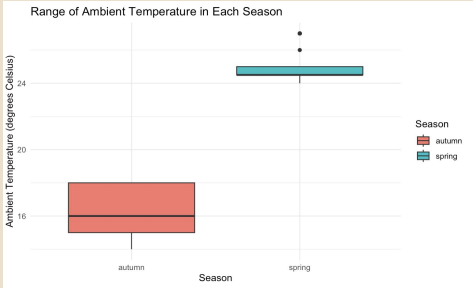
Leaf Size Range



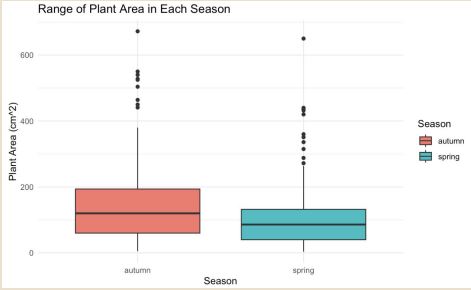
Plant Nitrogen Range



Ambient Temperature Range

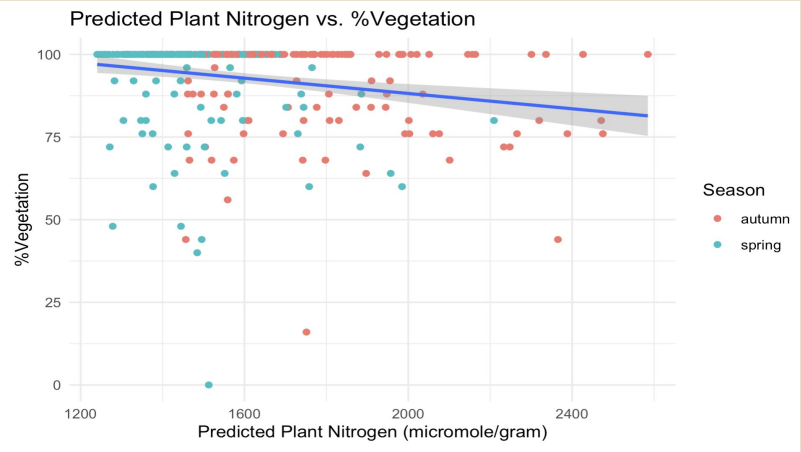


Plant Area Range

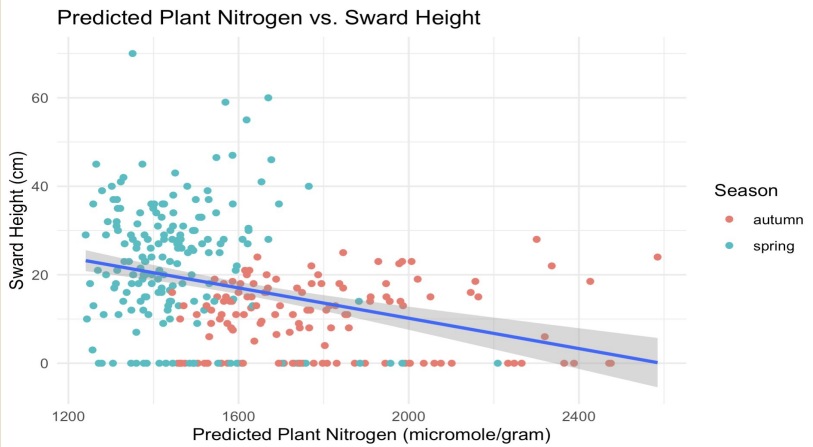


Plant nutrition and surrounding vegetation

Surrounding Vegetation

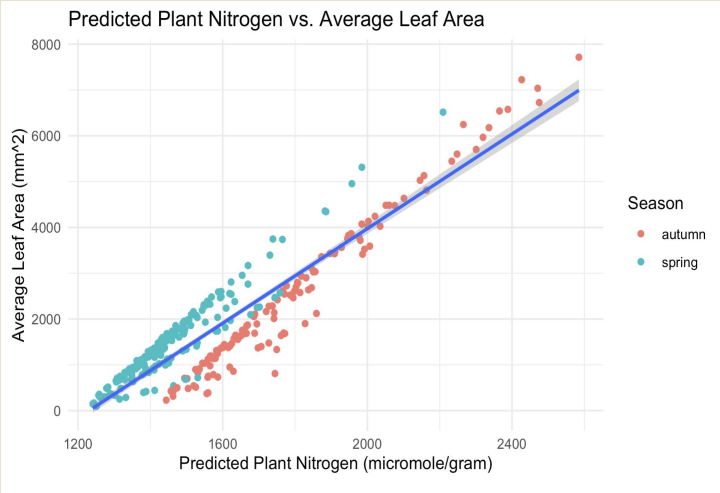


Sward Height

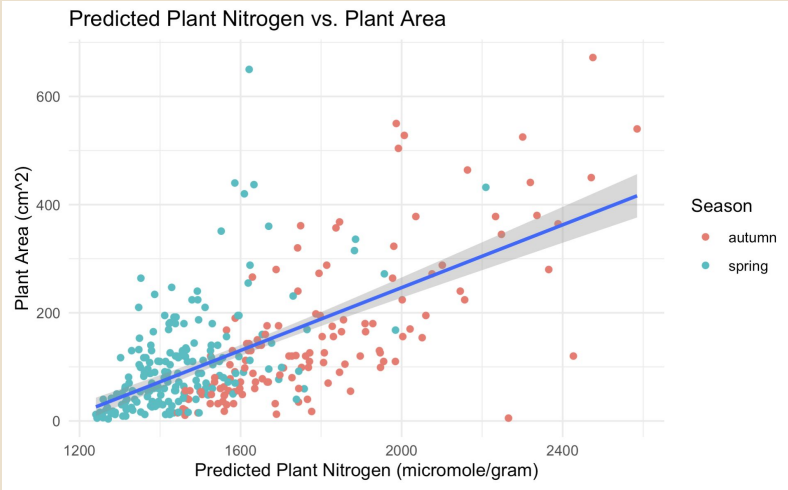


Leaf size and plant area as an indicator of plant nitrogen content

Average Leaf Size

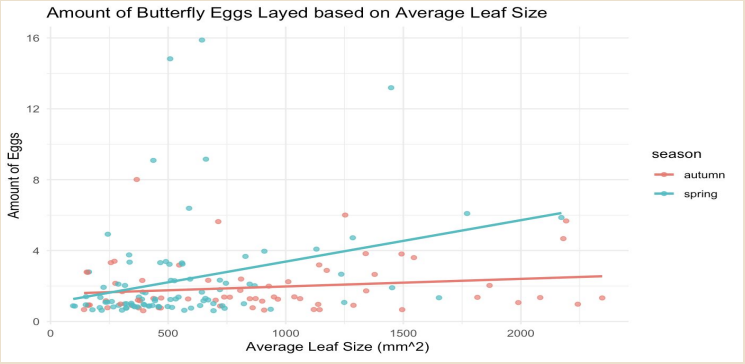


Plant Area

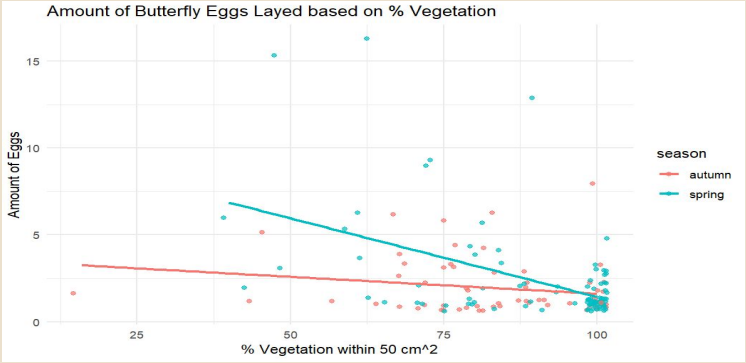


Egg Laying Preferences

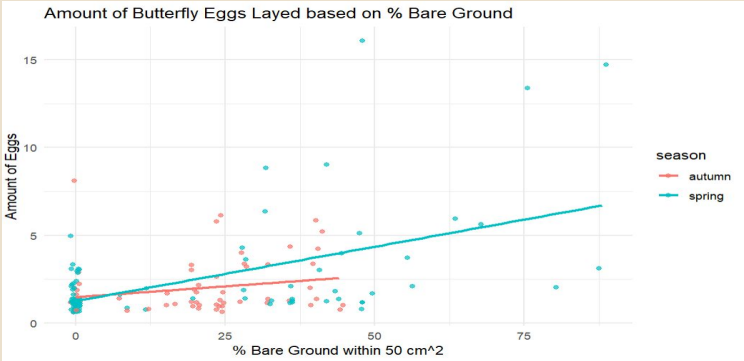
Average Leaf Size



% Vegetation

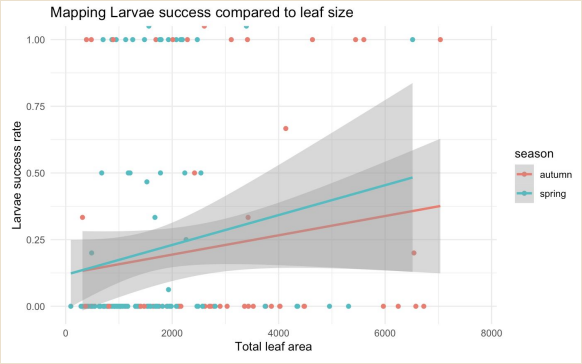


Soil Temperature

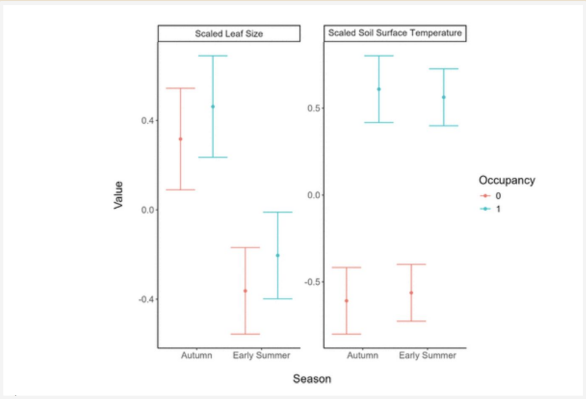


Egg laying success (larvae)

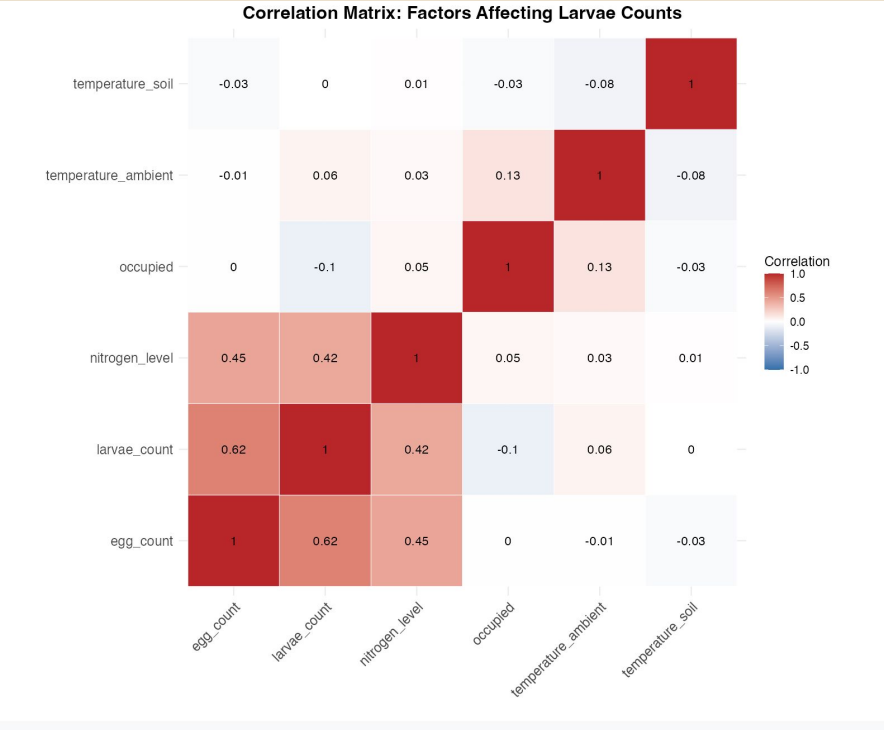
Leaf size influence



Seasonal effects



Correlation matrix



🦋 Butterfly Breeder App 🦋

Breeding Controls

🦋

Nitrogen Level Range:

0 — 5

0 0.5 1 1.5 2 2.5 3 3.5 4 4.5 5

Ambient Temperature (°C):

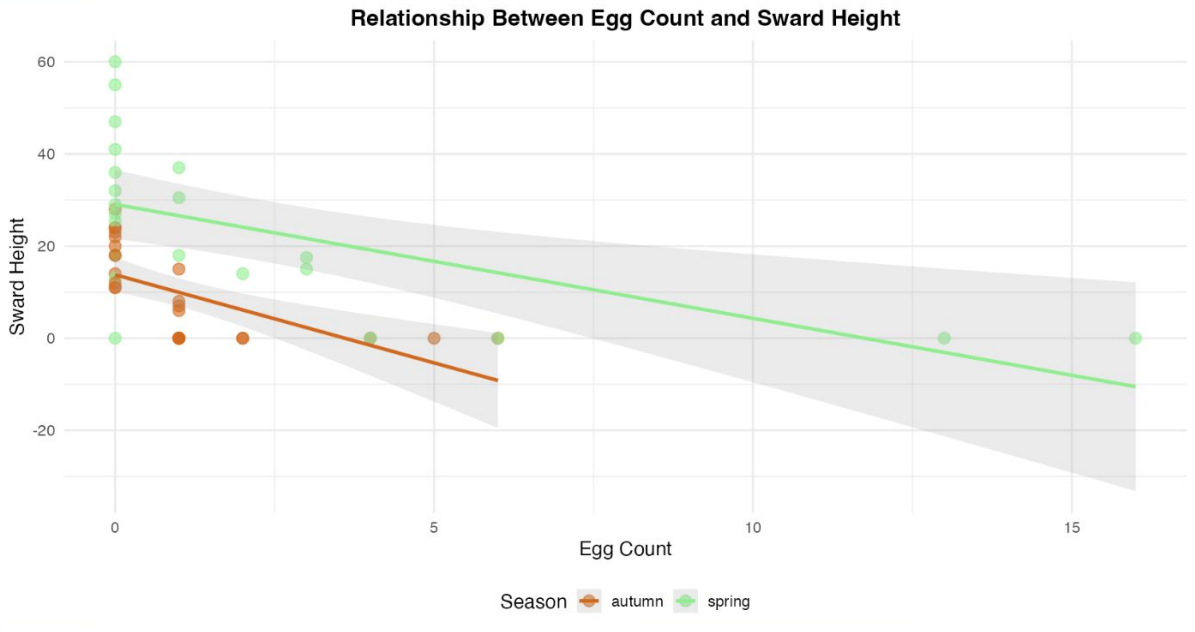
14 — 27

14 15.5 17 18.5 20 21.5 23 24.5 26 27

Select Seasons:

- [x] Spring
- [x] Autumn

Egg Count vs Sward Height by Season



Egg Count Distribution by Season

Larvae Count Distribution by Season

Egg Calculator Regression

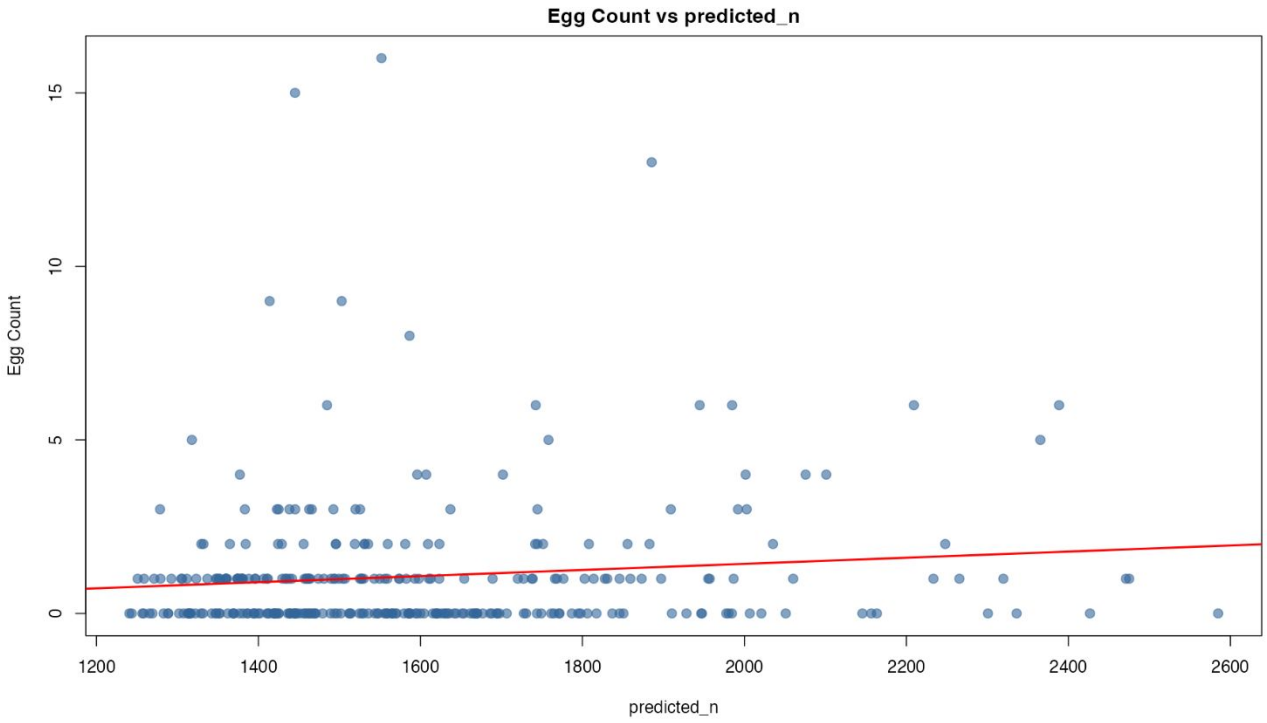
Select Predictor Variables:

- [x] predicted_n
- [ ] bare_ground
- [ ] veg
- [ ] sward_height
- [ ] plant_area
- [ ] average_leaf_size
- [ ] temperature_ambient
- [ ] temperature_soil

Run Regression

Scatter Plots | Regression Summary | Diagnostic Plots

Egg Count vs Predictor Variables



What did we conclude?

- Leaf size and plant size are both strong predictors of plant nutrition
- Butterflies seem to prefer laying eggs on plants with wider leaves, less surrounding vegetation, and more bare ground
- A limitation of this study would be the fact that the data was sampled in one environment, looking at specifically one plant and one butterfly species